

Question	Answer	Mark	Mark scheme	Additional guidance
17	56.0	P1	for a start to the process by correctly substituting into the cosine rule to find an angle eg, $18.2^2 = 14.6^2 + 7.9^2 - 2 \times 14.6 \times 7.9 \times \cos A$	$\cos B = 0.9069\dots$ $B = 24.912\dots$ $\cos C = 0.6276\dots$ $C = 51.1222\dots$ [A], [B], [C] must be a numerical value and clearly identified by labelling or on the diagram with no contradiction
		P1	for rearranging to find $\cos A$, eg $\cos A = \frac{14.6^2 + 7.9^2 - 18.2^2}{2 \times 14.6 \times 7.9}$ ($= -0.2413\dots$) or $A = 103.965\dots$	
		P1	for process to find the area of triangle ABC eg $\text{Area} = \frac{1}{2} \times 7.9 \times 14.6 \times \sin ("103.965\dots")$ or $\text{Area} = \frac{1}{2} \times 7.9 \times 14.6 \times \sin [A]$ $\text{Area} = \frac{1}{2} \times 14.6 \times 18.2 \times \sin ("24.912\dots")$ or $\text{Area} = \frac{1}{2} \times 14.6 \times 18.2 \times \sin [B]$ $\text{Area} = \frac{1}{2} \times 7.9 \times 18.2 \times \sin ("51.1222\dots")$ or $\text{Area} = \frac{1}{2} \times 7.9 \times 18.2 \times \sin [C]$	
		A1	for answer in the range 55.96 to 56.0	If an answer is given in the range in working and then rounded incorrectly award full marks